CMPn332 – CMPN332 – CMP432

For the image matrix F(x,y) determines the following parameters:

```
Image Matrix =
[255, 210, 202, 205, 204, 201, 203, 206, 204, 219, 255,
 253,  42,   2,   1,   0,   4,   1,   0,   1,  67, 252,
 254,  49,   1,   0,   3,   0,   0,   2,   2,  70, 255,
 255,  51,   0,   0,   0,   3,   0,   0,   3,  72, 255,
 254,  47,   0,   6,   0,   0,   4,   0,   0,  75, 255,
 255,  54,   0,   0,   2,   0,   3,   0,   0,  75, 249,
 254,  40,   3,   1,   0,   3,   0,   0,   0,  65, 255,
 255,  85,  39,  45,  39,  45,  43,  44,  43, 106, 252,
```

- Dynamic range
- Spatial resolution
- Intensity resolution
- Find the 4-*adjacency*, 8-*adjacency and* m-*adjacency* giving the connectivity set $V = \{0,1,2,3,4,6,40\}$ for the pixel $f(2,2)$
- Using 4-adjacency and the connectivity set $V = \{0,1,2,3,4,6,40\}$, perform connected component labeling on the image matrix. Present the resulting labeled image matrix where each connected region is assigned a unique label, and background pixels (values not in V) are assigned label 0.

ANSWER:

- Dynamic range → 0:255
- Spatial resolution→8 x 11 =88 pixel
- Intensity resolution→ 8bits
- Find the 4-*adjacency*, 8-*adjacency and* m-*adjacency* giving the connectivity set $V = \{0,1,2,3,4,6,40\}$ for the pixel $f(2,2)$
    - Value at (2,2) = **1**, which ∈ V so,
    - **4-adjacency neighbors** of (2, 2) ∈ $V := \{f(1,2), f(3,2), f(2,3)\}$
    - **8–adjacency neighbors** of (2, 2) ∈ $V := \{f(1,2), f(3,2), f(2,3), f(1,3), f(3,3)\}$
    - **m-adjacency neighbors** of (2,2) ∈ $V := \{f(1,2), f(3,2), f(2,3)\}$is in N_4→automatically m-adjacent
        - : $N_D(2,2) \in V = \{f(1,3), f(3,3)$ ***check on connectivity***

- $f(2,2) N4 \in V \rightarrow \{f(1,2), f(3,2), f(2,3)\}$
- $f(1,3) N4 \in V \rightarrow \{f(1,2), f(1,4), f(2,3)\}$
- $N_4(2,2) \cap N_4(1,3) \in V = \{f(1,2), f(2,3)\}$
- $so\ it's\ not\ m\ adjacent$

- $f(2,2) N4 \in V \rightarrow \{f(1,2), f(3,2), f(2,3)\}$
- $f(3,3) N4 \in V \rightarrow \{f(3,2), f(3,4), f(2,3), f(4,3)\}$
- $N_4(2,2) \cap N_4(3,3) \in V = \{f(3,2), f(2,3)\}$
- $so\ it's\ not\ m\ adjacent$

- Using 4-adjacency and the connectivity set $V = \{0,1,2,3,4,6,40\}$

0	0	0	0	0	0	0	0	0	0	0
0	0	1	1	1	1	1	1	1	0	0
0	0	1	1	1	1	1	1	1	0	0
0	0	1	1	1	1	1	1	1	0	0
0	0	1	1	1	1	1	1	1	0	0
0	0	1	1	1	1	1	1	1	0	0
0	2	1	1	1	1	1	1	1	0	0
0	0	0	0	0	0	0	0	0	0	0

L1=L2

L1 = {F(1,2)…….F(6,8)}

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Define the following Term:

- Sampling: Digitizing the co-ordinate value.
- Quantization: Digitizing the amplitude value.
- Spatial resolution: is a measure of the smallest discernible detail in an image.
- Intensity resolution: refers to the smallest discernible change in intensity level.
- Dynamic range: describes the ratio between the maximum and minimum intensities in the Image.

Extract Image Dynamic Range:

```
#include<opencv2/opencv.hpp>
#include"iostream"
using namespace std;
using namespace cv;
int main()
{
	Mat src3 = imread("x.jpg",0);
	int max = 0, min=255 ;
	for (int i = 0; i < src3.rows; i++)
	{
		for (int j = 0; j < src3.cols; j++)
		{

			if (src3.at<uchar>(i, j) > max)
			{
				max = src3.at<uchar>(i, j);
			}
			if (src3.at<uchar>(i, j) < min)
			{
				min = src3.at<uchar>(i, j);
			}
		}
	}
	cout << "max_value=" << max << endl;
	cout << "min_value=" << min << endl;
}
```